

Below is the **Week 1 Java Coding Practice Worksheet**. It reviews familiar Python-style problems while helping Raina become comfortable with Java syntax, `Scanner`, variables, calculations, and basic `if/else` statements.

 Edit



Raina's Java Coding Practice

Week 1: Java Foundations

Recommended schedule: 5–6 days

Daily practice time: 30–45 minutes

Goal: Become comfortable writing, compiling, running, and debugging small Java programs.

Topics Practiced

- `public static void main`
- `System.out.println`
- Variables and data types
- `Scanner` input
- Arithmetic operators
- Integer division and remainder
- Basic `if`, `else if`, and `else`
- Converting familiar Python logic into Java

General Instructions

For each problem:

1. Create a separate Java file.
2. Use a meaningful class name.
3. Add a comment at the top with:
 - Your name
 - Date
 - Problem name
4. Ask the user for input when required.
5. Display clear output messages.
6. Test the program with at least three different inputs.

7. Do not copy solutions from the internet.

Example file header:

```
// Name: Raina  
// Date:  
// Problem: Personal Introduction
```

Day 1: Printing and Variables

Problem 1A: Personal Introduction

Create a program named:

PersonalIntroduction.java

Store the following information in variables:

- Your name
- Your grade
- Your school type
- Your favorite subject
- Your favorite programming language

Print the information as complete sentences.

Example Output

```
Hello! My name is Raina.  
I am in Grade 9.  
I attend a specialized high school.  
My favorite subject is mathematics.  
I am currently learning Java.
```

Requirements

Use at least:

- Two `String` variables
 - One `int` variable
 - Five print statements
-

Problem 1B: Robotics Team Profile

Create a program named:

`RoboticsProfile.java`

Create variables for:

- Team number
- Number of coding members
- Number of robots planned
- Whether you are the team lead
- Team name

Print a short robotics team profile.

Example Output

Team Name: Robo Falcons

Team Number: 3

Coding Members: 3

Robots Planned: 1

Team Lead: true

Reflection

Write one sentence as a Java comment explaining the difference between:

`String`

`int`

`boolean`

Day 2: Keyboard Input

Problem 2: Student Information Card

Create a program named:

`StudentInformation.java`

Use `Scanner` to ask the user for:

- Name
- Age

- Grade
- Favorite school subject
- Favorite extracurricular activity

Print a formatted student information card.

Example Interaction

```
Enter your name: Raina
Enter your age: 14
Enter your grade: 9
Enter your favorite subject: Math
Enter your favorite activity: Robotics
```

Example Output

```
-----
STUDENT INFORMATION
-----
Name: Raina
Age: 14
Grade: 9
Favorite Subject: Math
Favorite Activity: Robotics
-----
```

Testing Requirement

Run the program using information for three different imaginary students.

Debugging Challenge

What happens when you use `nextInt()` and then immediately use `nextLine()` ?

Add the necessary extra `nextLine()` statement so the program correctly reads all input.

Day 3: Arithmetic Practice

Problem 3A: School Supply Calculator

Create a program named:

`SchoolSupplyCalculator.java`

Ask the user for:

- Number of notebooks
- Price of one notebook
- Number of pens
- Price of one pen

Calculate:

- Total notebook cost
- Total pen cost
- Total purchase cost

Example Interaction

Number of notebooks: 4

Price of one notebook: 3.50

Number of pens: 6

Price of one pen: 1.25

Example Output

Notebook cost: \$14.0

Pen cost: \$7.5

Total cost: \$21.5

Requirements

Use:

- `int` for quantities
- `double` for prices
- Multiplication
- Addition

Problem 3B: Average of Three Scores

Ask the user for three test scores.

Calculate and display the average.

Example

```
First score: 88
Second score: 94
Third score: 91
Average score: 91.0
```

Important Test

Test the program with:

```
80, 81, 82
```

Make sure the result is displayed correctly as a decimal.

Thinking Question

Why might this calculation give an incorrect decimal result?

```
int average = (score1 + score2 + score3) / 3;
```

Write your answer as a comment.

Day 4: Division and Remainder

Problem 4A: Minutes Converter

Create a program named:

```
MinutesConverter.java
```

Ask the user to enter a total number of minutes.

Convert the number into:

- Hours
- Remaining minutes

Example 1

```
Enter total minutes: 135
135 minutes equals 2 hours and 15 minutes.
```

Example 2

Enter total minutes: 60
60 minutes equals 1 hour and 0 minutes.

Hint

Use:

/
%

Problem 4B: Team Grouping

The robotics club has a certain number of students. Each team can have three students.

Ask the user for the total number of students.

Display:

- Number of complete teams
- Number of students left over

Example

Enter the number of students: 17
Complete teams: 5
Students remaining: 2

Test Values

Test with:

12
13
17
21

Day 5: Decisions with If/Else

Problem 5A: Positive, Negative, or Zero

Create a program named:

NumberClassifier.java

Ask the user to enter an integer.

Display whether the number is:

- Positive
- Negative
- Zero

Example

```
Enter a number: -8
```

```
-8 is negative.
```

Problem 5B: Even or Odd

Extend the same program.

When the number is not zero, also display whether it is even or odd.

Example 1

```
Enter a number: 14
```

```
14 is positive.
```

```
14 is even.
```

Example 2

```
Enter a number: -7
```

```
-7 is negative.
```

```
-7 is odd.
```

Testing Requirement

Test with:

```
0
```

```
8
```

```
-8
```

```
7
```

```
-7
```

Day 6: Weekly Challenge

Problem 6: Study Session Evaluator

Create a program named:

```
StudySessionEvaluator.java
```

Raina wants to evaluate her daily Java practice.

Ask the user for:

- Number of minutes studied
- Number of problems attempted
- Number of problems completed

Calculate the completion percentage:

```
completed problems ÷ attempted problems × 100
```

Then display a study rating using the following rules:

- **Excellent session:** 40 or more minutes and at least 80% completed
- **Good session:** 30 or more minutes and at least 60% completed
- **Keep practicing:** Anything below those levels

Example 1

```
Minutes studied: 45
```

```
Problems attempted: 5
```

```
Problems completed: 4
```

```
Completion percentage: 80.0%
```

```
Rating: Excellent session!
```

Example 2

```
Minutes studied: 35
```

```
Problems attempted: 5
```

```
Problems completed: 3
```

```
Completion percentage: 60.0%
```

```
Rating: Good session!
```

Example 3

Minutes studied: 20

Problems attempted: 4

Problems completed: 2

Completion percentage: 50.0%

Rating: Keep practicing!

Important Requirement

Make sure decimal division is used when calculating the percentage.

Safety Check

What might happen when the number of attempted problems is zero?

Add an `if` statement that displays:

Completion percentage cannot be calculated because no problem

Optional Bonus Challenge

Temperature Converter

Create a program that asks for a temperature in Fahrenheit and converts it to Celsius.

Use this formula:

$$\text{Celsius} = (\text{Fahrenheit} - 32) \times 5 \div 9$$

Example

Enter temperature in Fahrenheit: 86

Temperature in Celsius: 30.0

Then display:

- `Cold` when Celsius is below 10
- `Comfortable` when Celsius is from 10 through 25
- `Hot` when Celsius is above 25

Week 1 Completion Checklist

Mark each item after completing it.

- ☐ I created a Java file for every main problem.
 - ☐ I used meaningful variable names.
 - ☐ I used Scanner to receive user input.
 - ☐ I used int, double, String, and boolean.
 - ☐ I practiced division and remainder.
 - ☐ I used if, else if, and else.
 - ☐ I tested every program with at least three inputs.
 - ☐ I corrected compiler errors myself before asking for help.
 - ☐ I can explain integer division.
 - ☐ I can explain the difference between = and ==.
-

End-of-Week Reflection

Answer these questions in two or three sentences each.

1. What part of Java was easier because you already knew Python?

2. What Java syntax was most confusing?

3. Which problem was the most interesting?

4. What error did you encounter most often?

5. Rate your Java confidence after Week 1.

1 2 3 4 5
Low High

For Week 1, the most important problems are **Student Information Card, Minutes Converter, Number Classifier, and Study Session Evaluator**. The smaller exercises build Java syntax confidence without introducing difficult algorithms too early.